

GEOMETRY

Friday, June 21, 2024 — 9:15 a.m. to 12:15 p.m., only

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 35 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice ...

A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

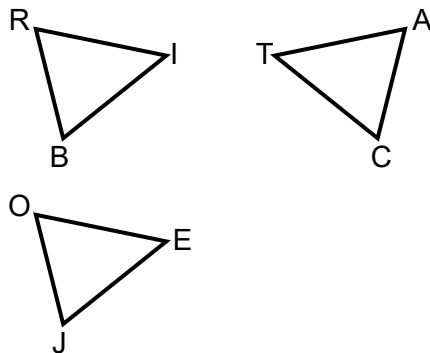
Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t-t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

- 1 In the diagram below, $\triangle BRI$ is the image of $\triangle JOE$ after a translation.
Triangle CAT is the image of $\triangle BRI$ after a line reflection.

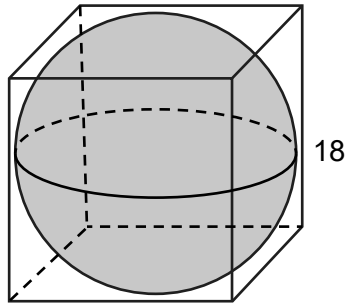


Which statement is always true?

- | | |
|-------------------------------|---|
| (1) $\angle R \cong \angle T$ | (3) $\overline{JE} \cong \overline{RI}$ |
| (2) $\angle J \cong \angle A$ | (4) $\overline{OE} \cong \overline{AT}$ |
- 2 A right cylinder is cut parallel to its base. The shape of this cross section is a
- | | |
|------------|---------------|
| (1) cone | (3) triangle |
| (2) circle | (4) rectangle |
- 3 What is the minimum number of degrees that a regular hexagon must rotate about its center to carry it onto itself?
- | | |
|----------------|-----------------|
| (1) 45° | (3) 60° |
| (2) 72° | (4) 120° |

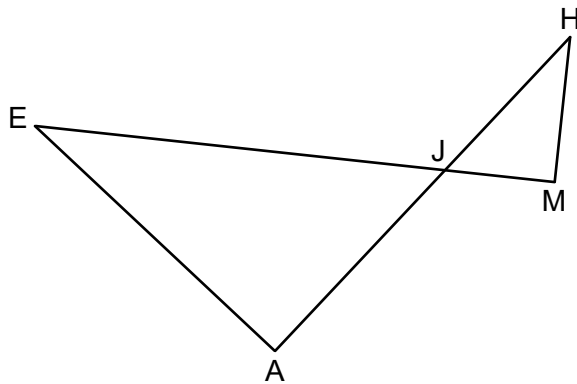
Use this space for
computations.

- 4 In the diagram below, a sphere is inscribed inside a cube. The cube has edge lengths of 18.



What is the volume of the sphere, in terms of π ?

- (1) 108π (3) 972π
(2) 432π (4) 7776π
- 5 In the diagram below, $\overline{EM}$ intersects $\overline{HA}$ at J , $\overline{EA} \perp \overline{HA}$, and $\overline{EM} \perp \overline{HM}$.



If $EA = 7.2$, $EJ = 9$, $AJ = 5.4$, and $HM = 3.29$, what is the length of $\overline{MJ}$, to the nearest hundredth?

- (1) 2.47 (3) 4.11
(2) 2.63 (4) 4.39

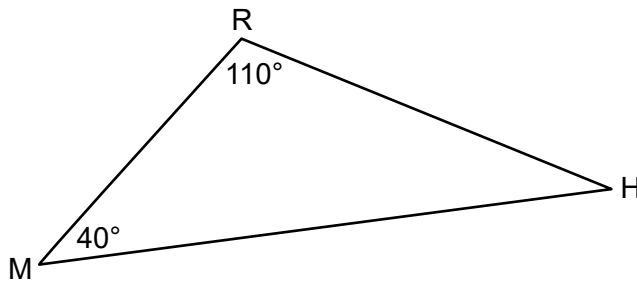
Use this space for
computations.

6 Which equation represents the line that passes through the point $(2, -7)$ and is perpendicular to the line whose equation is $y = \frac{3}{4}x + 4$?

(1) $y + 7 = \frac{3}{4}(x - 2)$ (3) $y + 7 = -\frac{4}{3}(x - 2)$

(2) $y - 7 = \frac{3}{4}(x + 2)$ (4) $y - 7 = -\frac{4}{3}(x + 2)$

7 In $\triangle RHM$ below, $m\angle R = 110^\circ$ and $m\angle M = 40^\circ$.

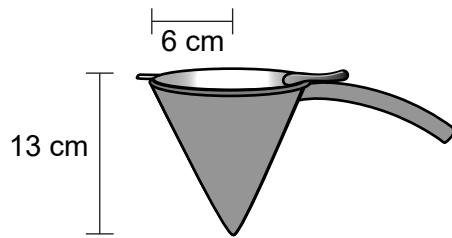


If $\triangle RHM$ is reflected over side $\overline{HM}$ to form quadrilateral $RHR'M$, which statement is always true?

- (1) Quadrilateral $RHR'M$ is a parallelogram.
- (2) $m\angle MHR' = 40^\circ$
- (3) $m\angle HMR' = 40^\circ$
- (4) $\overline{MR} \cong \overline{HR'}$

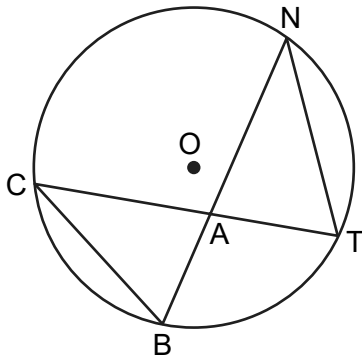
Use this space for
computations.

- 8 The funnel shown below can be used to decorate cookies with melted chocolate. The funnel can be modeled by a cone whose radius is 6 cm and height is 13 cm.



The baker uses 2 cubic centimeters of chocolate to decorate each cookie. When the funnel is completely filled, what is the maximum number of cookies that can be decorated with the melted chocolate?

- (1) 78 (3) 490
(2) 245 (4) 735
- 9 In circle O below, chords $\overline{CT}$ and $\overline{BN}$ intersect at point A . Chords $\overline{CB}$ and $\overline{NT}$ are drawn.

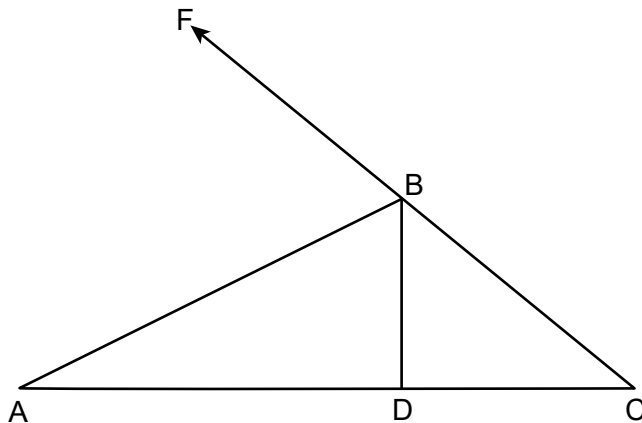


Which statement is always true?

- (1) $\frac{NT}{TA} = \frac{CB}{BA}$ (3) $\frac{NA}{AB} = \frac{TA}{AC}$
(2) $\angle BAC \cong \angle ATN$ (4) $\angle BCA \cong \angle NTA$

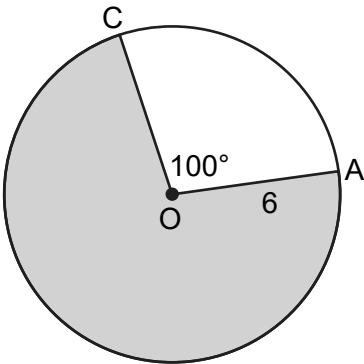
Use this space for
computations.

- 10 In the diagram below of $\triangle ABC$, $\overrightarrow{CBF}$ is drawn, $\overline{AB}$ bisects $\angle FBD$, and $\overline{BD} \perp \overline{AC}$.



If $m\angle C = 42^\circ$, what is $m\angle A$?

- (1) 24° (3) 48°
(2) 33° (4) 66°
- 11 In circle O below, $OA = 6$, and $m\angle COA = 100^\circ$.



What is the area of the shaded sector?

- (1) 10π (3) $\frac{10\pi}{3}$
(2) 26π (4) $\frac{26\pi}{3}$

**Use this space for
computations.**

12 In rectangle $ABCD$, diagonal $\overline{AC}$ is drawn. The measure of $\angle ACD$ is 37° and the length of $\overline{BC}$ is 7.6 cm. What is the length of $\overline{AC}$, to the *nearest tenth of a centimeter*?

- (1) 4.6
- (2) 9.5
- (3) 10.1
- (4) 12.6

13 A peanut butter manufacturer would like to use a cylindrical jar with a volume of 1180 cm^3 . The jar has a height of 10 cm. What is the diameter of the jar, to the *nearest tenth of a centimeter*?

- (1) 3.8
- (2) 6.1
- (3) 10.9
- (4) 12.3

14 Triangle KLM is dilated by a scale factor of 3 to map onto triangle DRS . Which statement is *not* always true?

- (1) $\angle K \cong \angle D$
- (2) $KM = \frac{1}{3}DS$
- (3) The area of $\triangle DRS$ is 3 times the area of $\triangle KLM$.
- (4) The perimeter of $\triangle DRS$ is 3 times the perimeter of $\triangle KLM$.

**Use this space for
computations.**

15 A rectangle with dimensions of 4 feet by 7 feet is continuously rotated about one of its 4-foot sides. The resulting three-dimensional object is a

- (1) cylinder with a height of 7 feet and a base radius of 4 feet.
- (2) cylinder with a height of 4 feet and a base radius of 7 feet.
- (3) cone with a height of 7 feet and a base radius of 7 feet.
- (4) cone with a height of 4 feet and a base radius of 7 feet.

16 In right triangle ABC , altitude $\overline{CD}$ is drawn to hypotenuse $\overline{AB}$. If $AD = 4$ and $CD = 8$, the length of $\overline{BD}$ is

- (1) $\sqrt{48}$
- (2) $\sqrt{80}$
- (3) 12
- (4) 16

17 If $ABCD$ is a parallelogram, which additional information is sufficient to prove that $ABCD$ is a rectangle?

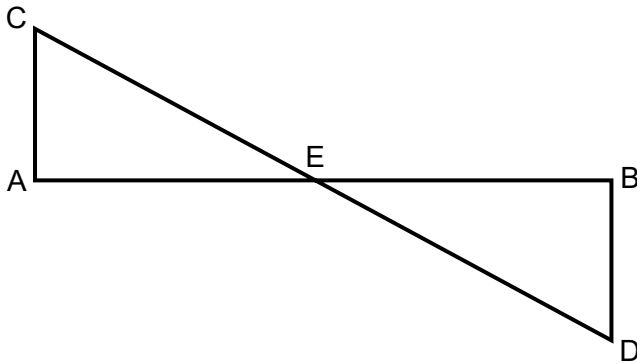
- (1) $\overline{AB} \cong \overline{BC}$
- (2) $\overline{AB} \parallel \overline{CD}$
- (3) $\overline{AC} \cong \overline{BD}$
- (4) $\overline{AC} \perp \overline{BD}$

Use this space for
computations.

18 Line segment APB has endpoints $A(-5,4)$ and $B(7,-4)$. What are the coordinates of P if $AP:PB$ is in the ratio 1:3?

- (1) $(-2,2)$ (3) $(1,0)$
(2) $(-1,1.3)$ (4) $(4,-2)$

19 In the diagram below, $\overline{AB}$ and $\overline{CD}$ intersect at E , and $\overline{CA}$ and $\overline{DB}$ are drawn.



If $\overline{CA} \parallel \overline{DB}$, which statement is always true?

- (1) $\overline{AE} \cong \overline{BE}$ (3) $\triangle AEC \sim \triangle BED$
(2) $\overline{CA} \cong \overline{DB}$ (4) $\triangle AEC \cong \triangle BED$

Use this space for
computations.

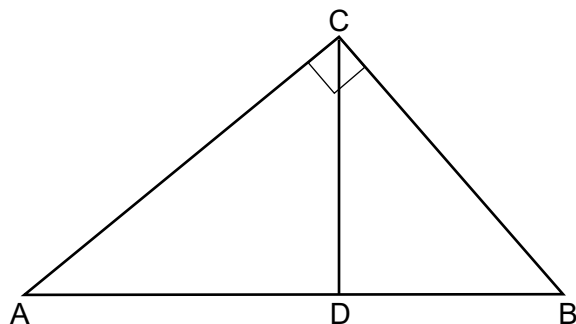
20 If $\sin(3x + 9)^\circ = \cos(5x - 7)^\circ$, what is the value of x ?

- (1) 8 (3) 33
(2) 11 (4) 42

21 Which set of integers could represent the lengths of the sides of an isosceles triangle?

- (1) {1, 1, 3} (3) {3, 3, 6}
(2) {2, 2, 5} (4) {4, 4, 7}

22 In the diagram shown below, altitude $\overline{CD}$ is drawn to the hypotenuse of right triangle ABC .



Which equation can always be used to find the length of $\overline{AC}$?

- (1) $\frac{AC}{CD} = \frac{CD}{AD}$ (3) $\frac{AC}{CD} = \frac{CD}{BC}$
(2) $\frac{CD}{AC} = \frac{AC}{AB}$ (4) $\frac{AB}{AC} = \frac{AC}{AD}$

**Use this space for
computations.**

23 Which congruence statement is sufficient to prove parallelogram *MARK* is a rhombus?

(1) $\overline{MA} \cong \overline{MK}$

(3) $\angle K \cong \angle A$

(2) $\overline{MA} \cong \overline{KR}$

(4) $\angle R \cong \angle A$

24 A line whose equation is $y = -2x + 3$ is dilated by a scale factor of 4 centered at (0,3). Which equation represents the image of the line after the dilation?

(1) $y = -2x + 3$

(3) $y = -8x + 3$

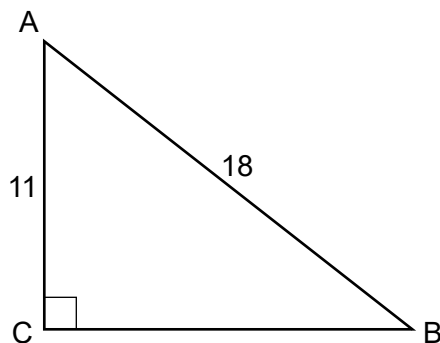
(2) $y = -2x + 12$

(4) $y = -8x + 12$

Part II

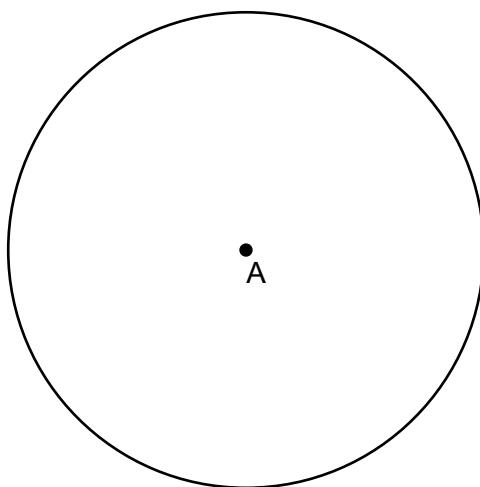
Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [14]

25 In $\triangle ABC$ below, $m\angle C = 90^\circ$, $AC = 11$, and $AB = 18$.

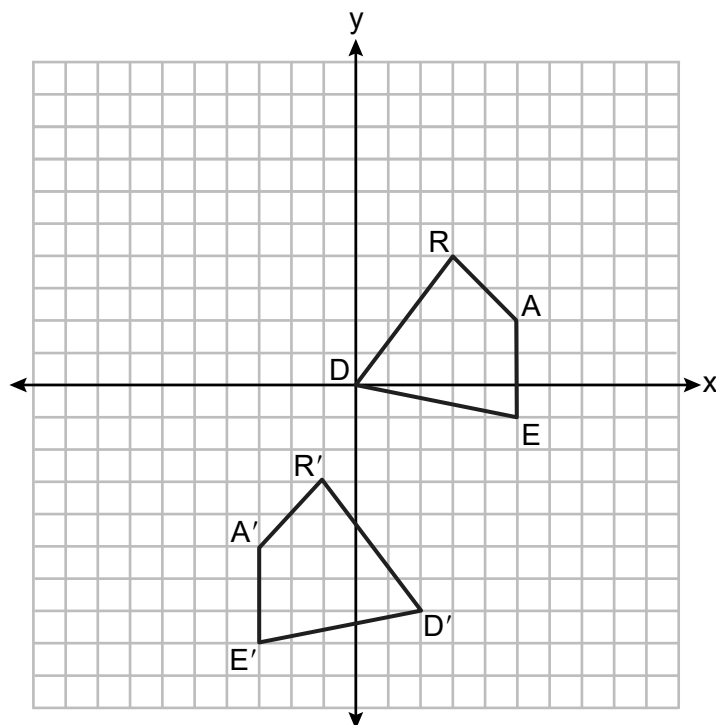


Determine and state the measure of angle A, to the *nearest degree*.

- 26** Use a compass and straightedge to construct an equilateral triangle inscribed in circle A below.
[Leave all construction marks.]

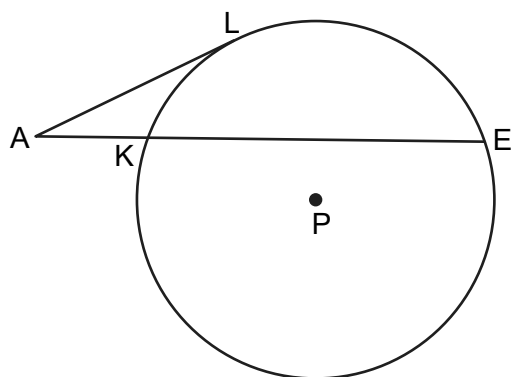


27 Quadrilateral $DEAR$ and its image, quadrilateral $D'E'A'R'$, are graphed on the set of axes below.



Describe a sequence of transformations that maps quadrilateral $DEAR$ onto quadrilateral $D'E'A'R'$.

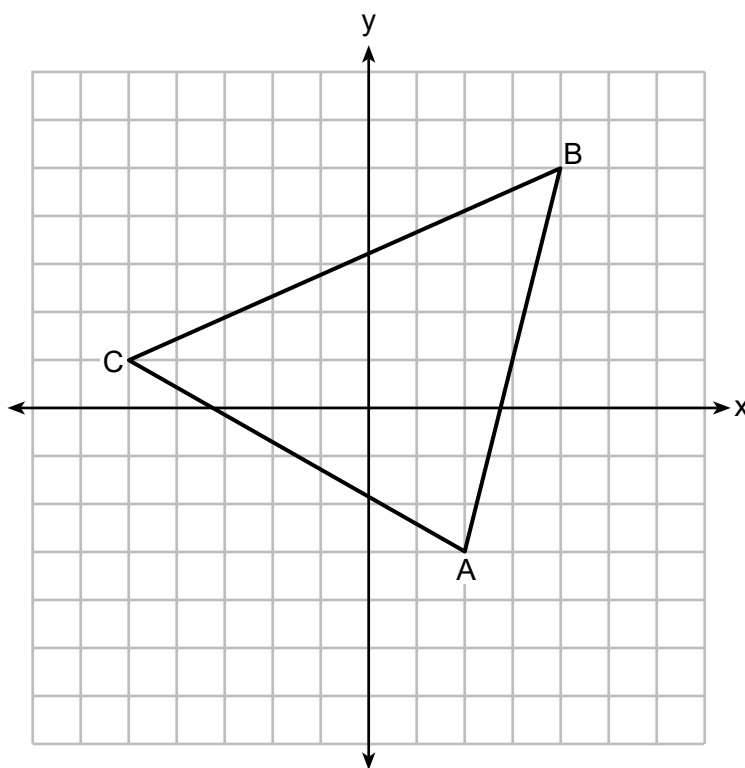
28 In circle P below, tangent $\overline{AL}$ and secant $\overline{AKE}$ are drawn.



If $AK = 12$ and $KE = 36$, determine and state the length of $\overline{AL}$.

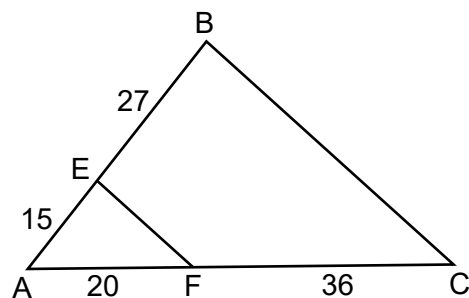
29 The equation of a circle is $x^2 + y^2 + 8x - 6y + 7 = 0$. Determine and state the coordinates of the center and the length of the radius of the circle.

- 30** On the set of axes below, $\triangle ABC$ is drawn with vertices that have coordinates $A(2, -3)$, $B(4, 5)$, and $C(-5, 1)$.



Determine and state the area of $\triangle ABC$.

31 In the diagram below, $AE = 15$, $EB = 27$, $AF = 20$, and $FC = 36$.

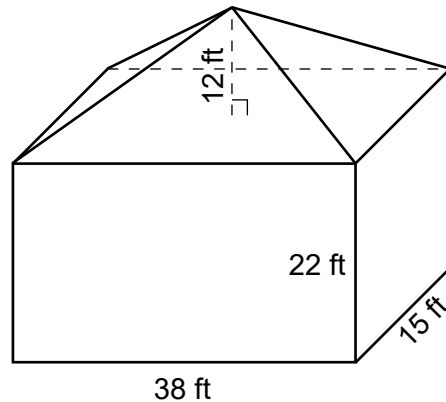


Explain why $\overline{EF} \parallel \overline{BC}$.

Part III

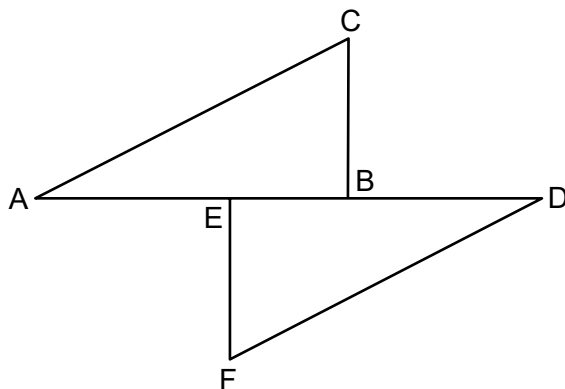
Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

- 32 A building is composed of a rectangular pyramid on top of a rectangular prism, as shown in the diagram below. The rectangular prism has a length of 38 feet, a width of 15 feet, and a height of 22 feet. The rectangular pyramid sits directly on top of the rectangular prism, and its height is 12 feet.



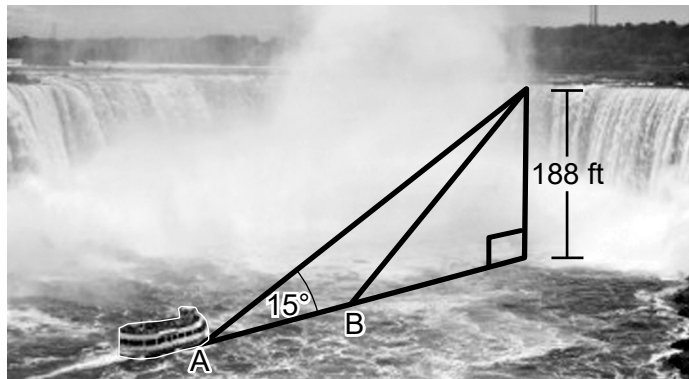
An air purification filter was installed that will clean all the air in the building at a rate of 2400 cubic feet per minute. Determine and state how long it will take, to the *nearest tenth of a minute*, for the filter to clean the air contained in the building.

33 Given: $\triangle ABC$, $\triangle DEF$, $\overline{AB} \perp \overline{BC}$, $\overline{DE} \perp \overline{EF}$, $\overline{AE} \cong \overline{DB}$, and $\overline{AC} \parallel \overline{FD}$



Prove: $\triangle ABC \cong \triangle DEF$

- 34** In the diagram below, a boat at point A is traveling toward the most powerful waterfall in North America, the Horseshoe Falls. The Horseshoe Falls has a vertical drop of 188 feet. The angle of elevation from point A to the top of the waterfall is 15° .



After the boat travels toward the falls, the angle of elevation at point B to the top of the waterfall is 23° . Determine and state, to the *nearest foot*, the distance the boat traveled from point A to point B .

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

35 Triangle JOE has vertices whose coordinates are $J(4,6)$, $O(-2,4)$, and $E(6,0)$.

Prove that $\triangle JOE$ is isosceles.

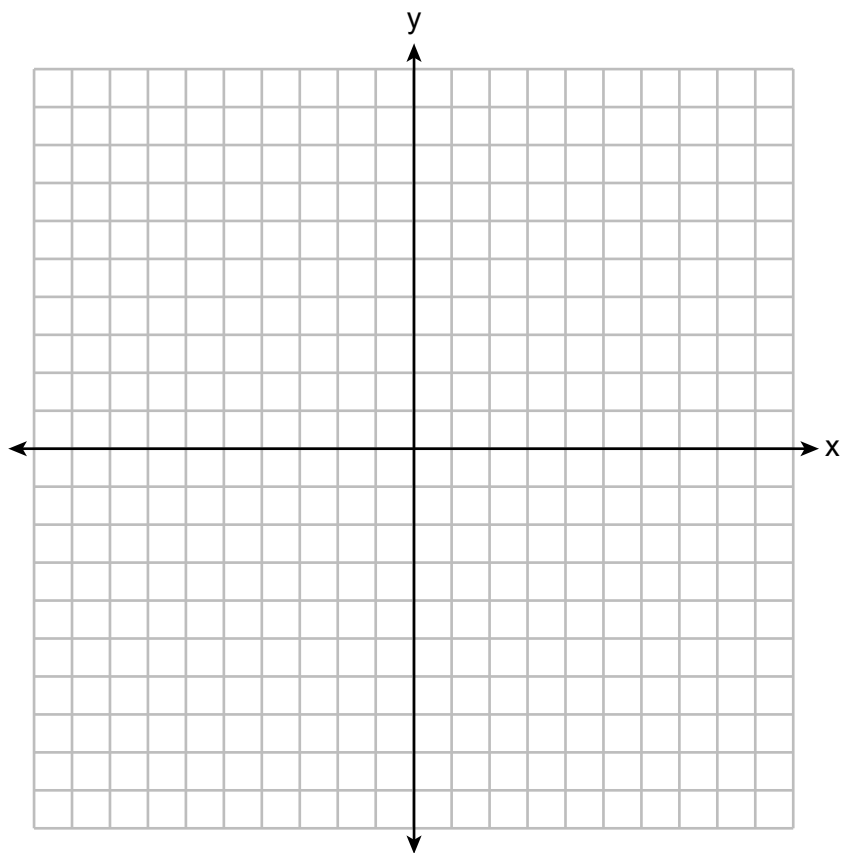
[The use of the set of axes on the next page is optional.]

Question 35 is continued on the next page.

Question 35 continued

Point $Y(2,2)$ is on $\overline{OE}$.

Prove that $\overline{JY}$ is the perpendicular bisector of $\overline{OE}$.



Regents Examination in Geometry – June 2024

Scoring Key: Part I (Multiple-Choice Questions)

Examination	Date	Question Number	Scoring Key	Question Type	Credit
Geometry	June '24	1	4	MC	2
Geometry	June '24	2	2	MC	2
Geometry	June '24	3	3	MC	2
Geometry	June '24	4	3	MC	2
Geometry	June '24	5	1	MC	2
Geometry	June '24	6	3	MC	2
Geometry	June '24	7	3	MC	2
Geometry	June '24	8	2	MC	2
Geometry	June '24	9	1	MC	2
Geometry	June '24	10	1	MC	2
Geometry	June '24	11	2	MC	2
Geometry	June '24	12	4	MC	2
Geometry	June '24	13	4	MC	2
Geometry	June '24	14	3	MC	2
Geometry	June '24	15	2	MC	2
Geometry	June '24	16	4	MC	2
Geometry	June '24	17	3	MC	2
Geometry	June '24	18	1	MC	2
Geometry	June '24	19	3	MC	2
Geometry	June '24	20	2	MC	2
Geometry	June '24	21	4	MC	2
Geometry	June '24	22	4	MC	2
Geometry	June '24	23	1	MC	2
Geometry	June '24	24	1	MC	2

Regents Examination in Geometry – June 2024

Scoring Key: Parts II, III, and IV (Constructed-Response Questions)

Examination	Date	Question Number	Scoring Key	Question Type	Credit
Geometry	June '24	25	-	CR	2
Geometry	June '24	26	-	CR	2
Geometry	June '24	27	-	CR	2
Geometry	June '24	28	-	CR	2
Geometry	June '24	29	-	CR	2
Geometry	June '24	30	-	CR	2
Geometry	June '24	31	-	CR	2
Geometry	June '24	32	-	CR	4
Geometry	June '24	33	-	CR	4
Geometry	June '24	34	-	CR	4
Geometry	June '24	35	-	CR	6

Key

MC = Multiple-choice question
CR = Constructed-response question

The chart for determining students' final examination scores for the **June 2024 Regents Examination in Geometry** will be posted on the Department's web site at: <https://www.nysedregents.org/geometryre/> on the day of the examination. Conversion charts provided for the previous administrations of the Regents Examination in Geometry must NOT be used to determine students' final scores for this administration.

Map to the Learning Standards
Geometry
June 2024

Question	Type	Credits	Cluster
1	Multiple Choice	2	G-CO.B
2	Multiple Choice	2	G-GMD.B
3	Multiple Choice	2	G-CO.A
4	Multiple Choice	2	G-GMD.A
5	Multiple Choice	2	G-SRT.B
6	Multiple Choice	2	G-GPE.B
7	Multiple Choice	2	G-CO.A
8	Multiple Choice	2	G-MG.A
9	Multiple Choice	2	G-SRT.B
10	Multiple Choice	2	G-CO.C
11	Multiple Choice	2	G-C.B
12	Multiple Choice	2	G-SRT.C
13	Multiple Choice	2	G-MG.A
14	Multiple Choice	2	G-SRT.A
15	Multiple Choice	2	G-GMD.B
16	Multiple Choice	2	G-SRT.B
17	Multiple Choice	2	G-CO.C
18	Multiple Choice	2	G-GPE.B
19	Multiple Choice	2	G-SRT.B
20	Multiple Choice	2	G-SRT.C
21	Multiple Choice	2	G-CO.C
22	Multiple Choice	2	G-SRT.B
23	Multiple Choice	2	G-CO.C
24	Multiple Choice	2	G-SRT.A
25	Constructed Response	2	G-SRT.C
26	Constructed Response	2	G-CO.D
27	Constructed Response	2	G-CO.A
28	Constructed Response	2	G-C.A
29	Constructed Response	2	G-GPE.A
30	Constructed Response	2	G-GPE.B
31	Constructed Response	2	G-SRT.B
32	Constructed Response	4	G-MG.A
33	Constructed Response	4	G-CO.C
34	Constructed Response	4	G-SRT.C
35	Constructed Response	6	G-GPE.B

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2] 52, and correct work is shown.

[1] Appropriate work is shown, but one computational or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] A correct relevant trigonometric equation is written, but no further correct work is shown.

or

[1] 52, but no work is shown.

[0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] A correct construction is drawn showing all appropriate arcs, and the equilateral triangle is drawn.

[1] Appropriate work is shown, but one construction error is made.

or

[1] A correct construction is drawn showing all appropriate arcs, but the equilateral triangle is not drawn.

[0] A drawing that is not an appropriate construction is shown.

or

[0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

- (27) [2] A correct sequence of transformations is written.
- [1] An appropriate sequence of transformations is written, but one computational or graphing error is made.
- or**
- [1] An appropriate sequence of transformations is written, but one conceptual error is made.
- or**
- [1] An appropriate sequence of transformations is written, but it is incomplete or partially correct.
- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.
- (28) [2] 24, and correct work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] A correct equation is written to find the length of $\overline{AL}$, but no further correct work is shown.
- or**
- [1] 24, but no work is shown.
- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

- (29) [2] $(-4,3)$, $\sqrt{18}$ or equivalent, and correct work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] Correct work is shown to find $(x + 4)^2 + (y - 3)^2 = 18$.
- or**
- [1] Correct work is shown to find $(-4,3)$.
- or**
- [1] Correct work is shown to find $\sqrt{18}$.
- or**
- [1] $(-4,3)$ and $\sqrt{18}$, but no work is shown.
- [0] $(-4,3)$ or $\sqrt{18}$, but no work is shown.
- or**
- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.
- (30) [2] 32, and correct work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] 32, but no work is shown.
- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

(31) [2] Correct work is shown, and a correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] A correct proportion is written, but the explanation is incomplete or partially correct.

[0] A correct proportion is written, but the explanation is missing or incorrect.

or

[0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (32) [4] 6.2, and correct work is shown.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- or***
- [3] Correct work is shown to find the total volume of the building, but no further correct work is shown.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or***
- [2] Correct work is shown to find the volume of the rectangular prism and the volume of the rectangular pyramid. No further correct work is shown.
- [1] Correct work is shown to find the volume of the rectangular prism or the volume of the rectangular pyramid. No further correct work is shown.
- or***
- [1] 6.2, but no work is shown.
- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

- (33) [4] A complete and correct proof that includes a concluding statement is written.
- [3] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or incorrect, or the concluding statement is missing.
- [2] A proof is written that demonstrates a good understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or incorrect.
- [1] Only one correct relevant statement and reason are written.
- [0] The “given” and/or the “prove” statements are written, but no further correct relevant statements are written.

or

- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.

- (34) [4] 259, and correct work is shown.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- or**
- [3] Correct work is shown to find the distance from the base of the waterfall to point A and the base of the waterfall to point B , but no further correct work is shown.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or**
- [2] Appropriate work is shown, but one conceptual error is made.
- or**
- [2] Correct work is shown to find the distance from the base of the waterfall to either point A or point B , but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.
- or**
- [1] At least one correct relevant trigonometric equation is written, but no further correct work is shown.
- or**
- [1] 259, but no work is shown.
- [0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.
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Part IV

For this question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (35) [6] Correct work is shown to prove $\triangle JOE$ is an isosceles triangle and $\overline{JY}$ is the perpendicular bisector of $\overline{OE}$. Correct concluding statements are written.
- [5] Appropriate work is shown, but one computational or graphing error is made.
- or**
- [5] Appropriate work is shown, but one concluding statement is missing or incorrect.
- [4] Appropriate work is shown, but two computational or graphing errors are made.
- or**
- [4] Appropriate work is shown, but one conceptual error is made in proving $\overline{JY}$ is the perpendicular bisector of $\overline{OE}$.
- or**
- [4] Appropriate work is shown, but two concluding statements are missing or incorrect.
- or**
- [4] Correct work is shown to prove $\overline{JY}$ is the perpendicular bisector of $\overline{OE}$ and correct concluding statements are written. No further correct work is shown.
- [3] Appropriate work is shown, but three or more computational or graphing errors are made.
- or**
- [3] Appropriate work is shown, but two or more computational or graphing errors are made, and one concluding statement is missing or incorrect.
- or**
- [3] Appropriate work is shown, but one conceptual error is made in proving $\overline{JY}$ is the perpendicular bisector of $\overline{OE}$ and one computational or graphing error is made.
- or**
- [3] Appropriate work is shown, but three concluding statements are missing or incorrect.
- [2] Correct work is shown to prove $\triangle JOE$ is an isosceles triangle, and a correct concluding statement is written. No further correct work is shown.
- or**

[2] Correct work is shown to prove $\overline{JY}$ and $\overline{OE}$ are perpendicular, and a correct concluding statement is written. No further correct work is shown.

or

[2] Correct work is shown to prove $\overline{JY}$ bisects $\overline{OE}$ and a correct concluding statement is written. No further correct work is shown.

[1] Correct work is shown to find the lengths of $\overline{JO}$ and $\overline{JE}$, but no further correct work is shown.

or

[1] Correct work is shown to find the slopes of $\overline{JY}$ and $\overline{OE}$, but no further correct work is shown.

or

[1] Correct work is shown to find the midpoint of $\overline{OE}$ or the lengths of $\overline{OY}$ and $\overline{EY}$, but no further correct work is shown.

[0] A zero response does not contain enough relevant course-level work to receive any credit, does not satisfy the criteria for one or more credits, or is a correct response that was obtained by an obviously incorrect procedure.
